

Ayan Hussain

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Summary

B.S. Data Science & Applications undergraduate (IIT Madras) with hands-on experience shipping production-grade **RAG and agentic AI systems** – multi-agent LLM pipelines, multimodal vision-based verification, and autonomous tool-using agents – on Python/FastAPI backends with containerized deployment. Comfortable across the AI engineering lifecycle: retrieval pipelines, model integration, async APIs, and end-to-end delivery.

Technical Skills

Languages: Python, SQL, Java | **Tools:** Git, Postman, Power BI, Jupyter, Excel Advanced

AI / LLM Engineering: LangChain, LangGraph, RAG Pipelines, Multi-Agent Orchestration, Prompt Engineering, Gemini API (Text & Vision), Azure AI Search (semantic retrieval), Azure Speech / Translator

Backend & APIs: FastAPI, Flask, Async SQLAlchemy 2, Pydantic v2, JWT Auth, REST API Design, SlowAPI Rate Limiting

Databases & Caching: PostgreSQL (Supabase), SQLite, Redis, Celery (task queues)

ML & Data: Scikit-Learn, XGBoost, LightGBM, Isolation Forest, Pandas, NumPy, Optuna

DevOps, Cloud & Frontend: Docker, Docker Compose, GitHub Actions, Linux/Unix, Hugging Face Spaces, Vercel, Railway.app, React, TypeScript, Vue 3, Tailwind CSS

Projects

EcoQuest India – AI-Verified Sustainability Platform

2026

React 19, TypeScript, FastAPI, Async SQLAlchemy 2, Gemini 2.5 Flash Vision, Supabase

GitHub

- Built a full-stack gamified platform (React 19 + TypeScript, FastAPI + async SQLAlchemy 2) with Supabase JWT auth, Postgres storage, and image CDN.
- Architected a multimodal verification pipeline with Gemini 2.5 Flash Vision to classify eco-action photo uploads (8 activity types), auto-approving above a 70% confidence threshold and routing low-confidence cases to human review.
- Engineered an anti-abuse layer combining perceptual-hash (pHash) duplicate detection with SlowAPI rate limiting and daily upload caps; containerized the stack with Docker Compose.

BharatBot – Multilingual Multi-Agent AI

2026

Python, FastAPI, Gemini, Azure AI Search/Speech/Translator, Docker

GitHub

- Selected among the Top 250 teams at Microsoft AI Unlocked Campus Edition 2026 – an invite-only hackathon for IITs, IIMs, ISB, XLRI, and NID.
- Architected a multi-agent system (Agriculture, Health, Legal) serving web and voice across 8 Indian languages with auto language detection – Gemini backbone, Azure Speech/Translator for STT-TTS, and Azure AI Search for grounded RAG retrieval.
- Containerized via Docker with health-check/logging endpoints; led a 5-person team to a production-ready deploy on Hugging Face.

Gemini WebGen Agent – Autonomous Web App Generator

2025

Python, FastAPI, Gemini 2.5 Flash, Docker, GitHub API

GitHub

- Built an autonomous LLM agent that receives task briefs via HTTP POST, generates complete single-file web apps, and auto-commits/deploys to GitHub Pages – zero manual intervention.
- Engineered a multi-round, tool-using pipeline with a Safe Mode: Round 1 scaffolds a fresh repo from scratch, Round 2+ applies surgical agent-planned updates, auto-rejecting outputs under 30% of original file size.
- Deployed on Hugging Face Spaces via Docker with async polling endpoints and live log streaming, supporting concurrent jobs via FastAPI background tasks.

LiveSports – Sports Event Management Platform

2026

Flask-RESTX, Vue 3, TypeScript, Groq LLaMA-3.3-70B, Razorpay

Client: Beyond Thoughts

- Backend developer and code reviewer on a full-stack sports event platform (Flask-RESTX + Vue 3/TypeScript) shipping 33 REST endpoints for client Beyond Thoughts.
- Built a RAG-based chatbot on Groq's LLaMA-3.3-70B with function calling for tool-driven, context-aware query handling across the platform.
- Integrated Razorpay payments with HMAC-SHA256 webhook verification and a wallet/referral-discount system; wrote/reviewed 65 pytest cases covering the API surface.

Education

IIT Madras

2023 – 2027

B.S. Data Science & Applications

CGPA: 7.9/10

Coursework: Machine Learning, Deep Learning, DBMS, Modern Application Development

Mary Lucas S&C (2019 – 2022): Class XII, ISC Board – 77% | Class X, ICSE Board – 78%

Certifications

Programming with GenAI (NPTEL, IISc Bangalore) | 5-Day AI Agents Intensive (Google & Kaggle) | SQL Advanced (HackerRank) | Power BI Dashboards (IIT Madras) | ML using Python (Simplilearn)